

JS Decision Making Statements

1. What is Decision Making Statements?

- a. **Decision-Making Statements** are used in programming to make decisions based on certain conditions.
- b. These statements allow the program to execute a block of code only if a specified condition is true or false.

2. Types of Decision Making Statements:

- a. There are four common types of Decision-Making Statements in most programming languages:
 - i. if-statement
 - ii. if-else statement
 - iii. else-if ladder
 - iv. switch statement

3. If-Statement:

- a. The if statement executes a block of code only when a specified condition is **true**.
- b. **Syntax** →

```
 if (condition) {  
    // Code to execute if the condition is true  
}
```

4. If-else Statement:

- a. The *if-else* statement adds an alternative block of code to execute if the condition is false.
- b. **Syntax** →

```
💡 if (condition) {  
    // Code to execute if the condition is true  
}  
else {  
    // Code to execute if the condition is false  
}
```

5. Else-if Ladder:

- a. The *if-else* ladder allows multiple conditions to be checked one after another.
- b. Once a true condition is found, its block is executed, and the rest are ignored.
- c. **Syntax** →

```
💡 if (condition1) {  
    // Code to execute if condition1 is true  
}  
else if (condition2) {  
    // Code to execute if condition2 is true  
}  
else if (condition3) {  
    // Code to execute if condition3 is true  
}  
else {  
    // Code to execute if none of the above conditions are true  
}
```

6. Switch Statement:

- a. The **switch** statement is used when you have multiple conditions to check for a **single variable** or **expression**.
- b. It is more structured and readable compared to an if-else ladder.
- c. **Syntax** →

```
 switch (expression) {  
  
    case value1:  
        // Code to execute if expression matches value1  
  
        break;  
  
    case value2:  
        // Code to execute if expression matches value2  
  
        break;  
  
    case value3:  
        // Code to execute if expression matches value3  
  
        break;  
  
    default:  
        // Code to execute if no cases match  
  
        break;  
}
```

- d. **break** in switch:
 - i. **break** is one of the keywords in JavaScript.
 - ii. Ends the current **case** and exits the **switch** block immediately.
 - iii. Commonly used to prevent accidental execution of multiple cases.